

Egor Marin, ML Engineer/Scientist

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SUMMARY

- ML Scientist with a PhD in Structural Biology and 8+ years of research experience
- built and managed ML/DL codebase with data and model registry in a team of scientists and engineers
- specialized in building generative and predictive models for antibody drug discovery and structural biology
- managed complete research data lifecycles, from cryo-EM/crystallography data processing (20TB+) to deploying ML-driven analysis tools
- Core Developer for MDAnalysis, contributing major features like a parallel processing backend (dask/multiprocessing)

EXPERIENCE

Machine Learning Scientist

ENPICOM B.V.

May 2024 — Present

Den Bosch, Netherlands

- develop and deploy generative and predictive models for antibody drug discovery using pytorch/lightning and custom discrete diffusion pipelines
- own the end-to-end ML lifecycle, from sourcing and cleaning biochemical data to building reproducible training pipelines with MLflow for experiment tracking
- manage codebase with data and training pipelines, ensuring regular code reviews and code reusability while maintaining modularity and compatibility of the models

Open-source software engineer

MDAnalysis via Google Summer of Code

May 2023 — September 2023

Remote

- designed and wrote backward-compatible parallelization with dask or multiprocessing for the core analysis library

PhD Researcher

University of Groningen

June 2021 — Dec 2023

Groningen, Netherlands

- led a research project as a corresponding author, applying regression-based active learning to accelerate ultra-large library docking, resulting in a publication in J. Chem. Inf. Model
- designed, deployed, and managed a high-throughput data processing infrastructure (CryoSPARC) for 15+ users, processing over 80 cryo-EM projects and 20TB of data
- developed a custom web server (ntfy-cryosparc) to provide personalized notifications, improving workflow efficiency for the research group
- solved 38+ protein structures using cryo-EM and X-ray crystallography, leading to publications in journals including Nature Communications and Science Advances

Junior Research Associate

Moscow Institute of Physics and Technology

March 2017 — September 2021

Moscow, Russia

- managed the end-to-end protein crystallography pipeline for multiple research groups, from sample preparation to synchrotron and XFEL data collection
- processed, refined, and analyzed crystallographic data, successfully determining and depositing 38 protein structures to the PDB
- contributed structural data and analysis that formed the basis for 16 peer-reviewed publications

SKILLS

Technology: python | rust | pytorch | jax | scipy | numpy | sklearn | polars | pydantic | mlflow | docker | AWS | k8s | SLURM

Domain: cryoEM | X-ray crystallography | biophysics | drug discovery | cheminformatics | antibodies

SELECTED PUBLICATIONS

For full list, see [google scholar](#).

Science Advances

July 2025

CryoRhodopsins: a comprehensive characterization of a new clade of microbial rhodopsins from cold environments

Pre-print on biorxiv <i>Structural basis for no retinal binding in flotillin-associated rhodopsins</i>	29 April 2025
Nature Communications <i>A subgroup of light-driven sodium pumps with an additional Schiff base counterion</i>	4 October 2024
Journal of Chemical Information and Modeling <i>Regression-Based Active Learning for Accessible Acceleration of Ultra-Large Library Docking</i>	December 2023
Nature Communications Chemistry <i>Mirror proteorhodopsins</i>	2 May 2023
Nature Communications <i>Structural basis for receptor selectivity and inverse agonism in S1P5 receptors</i>	August 2022
Biomolecules <i>Structure-Based Virtual Screening of Ultra-Large Library Yields Potent Antagonists for a Lipid GPCR</i>	December 2020
Structural basis of ligand selectivity and disease mutations in cysteinyl leukotriene receptors <i>Nature Communications</i>	December 2019
Science Advances <i>Structure-based mechanism of cysteinyl leukotriene receptor inhibition by antiasthmatic drugs</i>	October 2019

EDUCATION

University of Groningen <i>PhD</i>	2019 — 2023 <i>Groningen, Netherlands</i>
- thesis ‘On the methods of studying protein-ligand interaction dynamics’ - methods: cryoEM, X-ray crystallography, molecular dynamics, protein biochemistry, protein biophysics - publications in JACS, Crystal Growth & Design, Journal of Cheminformatics & Modelling	
Moscow Institute of Physics and Technology <i>MSc in applied mathematics and physics</i>	2017 — 2019 <i>Moscow, Russia</i>
graduated <i>summa cum laude</i>	
Moscow Institute of Physics and Technology <i>BSc in applied mathematics and physics</i>	2013 — 2017 <i>Moscow, Russia</i>
- graduated <i>magna cum laude</i> - related coursework: Calculus I-IV, Linear Algebra I-II, Complex Analysis, Differential Equations I-II, Analytical Mechanics I-II, Thermodynamics.	

OPEN SOURCE CONTRIBUTIONS & PROJECTS

Open-source contributions	Jan 2021 — Present
polars: contributed to polars (issue #25383: extending <code>replace_many</code> with <code>leftmost</code> option. polars-distance: minor contribution to polars plugin for distance calculation reciprocalspaceship: wrote parser for serial crystallography data into binary dataframe-like class	
MDAnalysis Core Developer	February 2025 — Present
MDAnalysis Core Developer - wrote a parallel backend for all analysis classes (dask/multiprocessing) - added a DSSP module for native secondary structure assignment - currently working on fast unified MMCIF parser based on gemmi	
Self-hosting	Jan 2021 — Present
self-hosting multiple applications (paperless, llama.cpp, openwebui) for family use under a Tailscale VPN	
Data Science Competitions	2018 — Present
- top-10% in Kaggle “Predict Molecular Properties” (public notebooks + gradient boosting on self-written rotationally invariant features) - top-10% in Takeda competition at Signate 5th place in Tochka Bank graph ML competition	